# Proteus: Drift-Aware, GAN-Augmented Intrusion Detection for Software-Defined Networks

Adhyan Jain
Independent Research

***Abstract*—Network intrusion detection systems (IDS) trained on static datasets degrade under concept drift, particularly in Software-Defined Networks (SDN) where traffic patterns and attack families evolve continuously. We present Proteus, an IDS architecture that combines an adaptive classifier with a conditional Wasserstein GAN with Gradient Penalty (WGAN-GP) for minority-class synthetic augmentation, a Kolmogorov-Smirnov (KS) test-based drift detector operating on classifier prediction-confidence distributions, and a Maximum Mean Discrepancy (MMD) fidelity gate that admits or rejects synthetic samples before adaptation. To avoid full-dataset retraining computational bottlenecks ($O(N)$ memory, multi-minute latency), we introduce a bounded reference-reservoir and sliding adaptation buffer (BOUNDEDBUFFERCLASSIFIER) guaranteeing $O(1)$ memory footprint and sub-3s adaptation latency per drift event. We evaluate three conditions – a frozen baseline, one-shot static augmentation, and a closed-loop drift-triggered adaptation pipeline – on the real CICIDS2017 dataset under a genuine temporal drift split (Monday–Thursday training vs. Friday holdout, which contains attack families entirely absent from training). At full scale (1.47M training rows), static augmentation did not improve macro-F1 over the baseline (0.8915 vs. 0.8922), an honest negative result. Under genuine temporal drift, closed-loop incremental adaptation achieved a final macro-F1 of 0.1738 (post-drift mean 0.1495) vs. 0.0294 for the baseline (a 5.91x relative macro-F1 ratio), evaluated strictly on a held-out 30% evaluation split of each stream window – computed once per window and shared identically across all three conditions – to eliminate train/test leakage and condition-to-condition evaluation asymmetry. A targeted ablation attributes this recovery to incremental real-data adaptation itself rather than to GAN/synthetic augmentation, whose measured role is defensive (see Section X). This is a single-seed result ($n = 1$); a statistically validated multi-seed evaluation is future work. We document the architecture, the audit finding that identified the leakage bug in prior evaluations, its unit-tested fix, the incremental adaptation mechanism, and the complete validated single-seed evaluation.**

***Index Terms*—intrusion detection, concept drift, generative adversarial networks, WGAN-GP, maximum mean discrepancy, software-defined networking, CICIDS2017**

## I. INTRODUCTION

Machine-learning-based network intrusion detection systems are typically trained once on a static labeled corpus and deployed without further adaptation. In practice, network traffic distributions drift: new attack tools appear, benign traffic patterns shift with application usage, and rare attack classes remain chronically under-represented in any single collection window. Software-Defined Networks (SDN) make this worse operationally (a compromised or misconfigured controller can reshape traffic at scale) while also making it more tractable to address (a centralized controller is a natural place to instrument feature export and retrain triggers).

Proteus targets this problem directly: rather than assuming a fixed training distribution, it instruments a live (or replayed) traffic stream with (1) a lightweight drift detector that requires no ground-truth drift labels, (2) a generative mechanism (WGAN-GP) that can synthesize additional minority-class examples once drift is detected, and (3) a fidelity gate that independently verifies a synthetic batch resembles real recent data before it is allowed to influence retraining. This paper documents the architecture as implemented, the datasets and experimental protocol actually used, and – in the spirit of reproducible, honest reporting – a real methodological bug found during a post-hoc audit of this same codebase, its fix, and exactly what evidence does and does not yet exist for the corrected pipeline.

### A. Contributions

- A concrete, open, full-scale (real CICIDS2017 + InSDN, GPU-trained) implementation of a closed-loop drift-detect $\rightarrow$ GAN-update $\rightarrow$ fidelity-gate $\rightarrow$ retrain pipeline, evaluated against a genuine temporal-drift split rather than synthetically injected drift.
- An MMD-based fidelity gate analysis showing a real, unresolved tension between two fidelity signals: a per-class diversity diagnostic (0/12 classes mode-collapsed) and a stricter distribution-level MMD comparison (only 44% of diversity-passing synthetic batches also pass the MMD gate).
- A documented, fixed train/test leakage bug in the closed-loop evaluation's post-adaptation metric, presented as a methodological contribution in its own right: the bug pattern (scoring a retrained model on the exact rows it was just trained on) is a plausible failure mode for any online/continual-learning evaluation and is not always obvious from reading the code once.

## II. RELATED WORK

**Intrusion detection with CICIDS2017.** Sharafaldin, Lashkari, and Ghorbani introduced the CICIDS2017 dataset as a modern, labeled, multi-day capture addressing known limitations (staleness, lack of traffic diversity) of older IDS benchmarks such as KDD-99 and NSL-KDD [1]. It has since

become a standard benchmark for supervised network IDS work, though the well-documented long-tailed class distribution (single-digit sample counts for some attack subtypes) remains a challenge any evaluation on it must address explicitly rather than average over.

**Generative Adversarial Networks for minority-class augmentation.** Goodfellow et al. introduced the GAN framework as an adversarial minimax game between a generator and discriminator [2]. Standard GAN training is well known to be unstable; Arjovsky, Chintala, and Bottou reformulated the objective using the Wasserstein distance to improve training stability [3], and Gulrajani et al. introduced the gradient penalty (WGAN-GP) as a more stable alternative to weight clipping for encouraging the critic toward the desired Lipschitz constraint the Wasserstein objective requires [4]. Proteus's GAN component is a conditional WGAN-GP built directly on this line of work, used specifically for minority-class attack-sample synthesis rather than general-purpose image or tabular generation.

**Distributional similarity via Maximum Mean Discrepancy.** Gretton et al. formalized MMD as a kernel-based two-sample test statistic for determining whether two samples are drawn from the same distribution [5]. Proteus uses a Gaussian-kernel MMD estimate, with the kernel bandwidth chosen by the median heuristic, as the core of its fidelity gate: a synthetic batch is admitted only if its MMD to a real reference batch falls below a fixed threshold.

**Concept drift detection.** A large literature addresses drift detection in streaming data; Proteus takes a specific, simple approach – rather than monitoring the raw feature distribution, it monitors the classifier's own prediction-confidence distribution over time via a two-sample Kolmogorov-Smirnov test against a fixed reference window. This requires no ground-truth drift labels and directly measures what matters operationally: whether the deployed model's own confidence calibration has shifted.

**Random Forests.** Breiman's Random Forest [6] is used throughout as the IDS classifier itself, chosen for its robustness to feature scaling differences and its native handling of multi-class, class-imbalanced tabular data without architecture search.

**InSDN.** Elsayed et al. introduced InSDN as an SDN-specific intrusion dataset captured from an emulated SDN testbed [7], used in this project as a second, SDN-native data source alongside CICIDS2017 (see Section VI).

## III. Research Questions

- **RQ1**: Does offline, one-shot GAN-based synthetic augmentation improve a Random-Forest IDS's macro-F1 on a fixed test set, at full real-data scale?
- **RQ2**: Under genuine temporal distribution drift (unseen attack families arriving after training), does a closed-loop drift-detect/retrain pipeline recover classification performance better than a frozen baseline or a one-shot-augmented classifier?
- **RQ3**: Does an MMD-based fidelity gate reliably distinguish real generator output from injected noise, and how does its admission behavior compare against a simpler per-class diversity diagnostic?

RQ1 and RQ3 have real, executed evidence at full scale (Sections VII-A and VII-B). RQ2 – the project's central hypothesis – does not currently have valid full-scale evidence; Section VII-C explains why and what is required to obtain it.

## IV. Proteus Architecture

The end-to-end pipeline, identical in structure across the small-scale demo implementation (`proteus/{data,baseline,gan,drift,fidelity,stream,p` and the full-scale research implementation (`proteus/{data_full,baseline_full,gan_full,closed_loop_full,evaluate_full}.py`), is:

*data → preprocessing → baseline IDS → stream → drift detection → GAN update → synthetic generation → fidelity gate → admission → classifier retraining → continued evaluation*

### A. Baseline classifier

A Random Forest (100 trees, fixed random seed) trained on the labeled training pool. Evaluated via macro-F1, weighted-F1, and the full per-class classification report.

### B. Conditional WGAN-GP

A conditional generator/critic pair trained per-class on the subset of training rows for each GAN-target class. Target classes are selected by two criteria, both required: (1) rarity (below 3% of the training corpus) and (2) modelability (at least 300 training rows) – a class with a single-digit sample count cannot be meaningfully modeled or validated against, and augmenting it would constitute fabrication rather than augmentation. The critic is trained with the WGAN-GP objective:

$$\mathcal{L}_D = \mathbb{E}_{\tilde{x}\sim\mathbb{P}_g}[D(\tilde{x})] - \mathbb{E}_{x\sim\mathbb{P}_r}[D(x)] + \lambda\,\mathbb{E}_{\hat{x}\sim\mathbb{P}_{\hat{x}}}\left[(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2\right] \tag{1}$$

where $\hat{x}$ is sampled uniformly along straight lines between real and generated pairs, per [4].

### C. Drift detector

Given a reference window of classifier prediction-confidence scores $C_{\text{ref}}$ and a current window $C_{\text{cur}}$, the detector computes a two-sample KS statistic and declares drift fired if the resulting $p$-value falls below $\alpha = 0.05$:

$$D_{n,m} = \sup_x |F_{\text{cur}}(x) - F_{\text{ref}}(x)| \tag{2}$$

### D. Fidelity gate

Given a synthetic batch $X$ and a real reference batch $Y$, an unbiased Gaussian-kernel MMD estimate is computed with bandwidth set via the median heuristic:

$$\widehat{\text{MMD}}^2(X,Y) = \frac{1}{n^2}\sum_{i,j} k(x_i,x_j) + \frac{1}{m^2}\sum_{i,j} k(y_i,y_j) - \frac{2}{nm}\sum_{i,j} k(x_i,y_j) \tag{3}$$

A batch is admitted iff $\widehat{\text{MMD}}^2 < \tau$ (fixed at $\tau = 0.25$ throughout).

**Algorithm 1** ClosedLoopOrchestrator.step (post-fix)

$X_{\text{eval}}, y_{\text{eval}}, X_{\text{fit}}, y_{\text{fit}} \leftarrow \text{SPLITEVALFIT}(X, y, 0.3)$
$f1_{\text{before}} \leftarrow \text{MacroF1}(clf, X_{\text{eval}}, y_{\text{eval}})$
$conf \leftarrow clf.\text{predict_proba}(X).\max(\text{axis} = 1)$
$result \leftarrow \text{DETECTDRIFT}(conf, ref_conf)$
**if** $result$.fired **then**
    $\text{RESUMEGAN}(X_{\text{fit}}, y_{\text{fit}})$
    **for** each admit-eligible class $c$ **do**
        $synth \leftarrow \text{GENERATE}(c)$
        $admitted, score \leftarrow \text{FIDELITY-GATE}(synth, X_{\text{fit}}[y_{\text{fit}} = c])$
        **if** $admitted$ **then**: append $synth$ to training pool
        **end if**
    **end for**
    $X_{\text{train}} \leftarrow X_{\text{train}} \cup X_{\text{fit}} \cup \{\text{admitted}\}$
    retrain $clf$ on $X_{\text{train}}$
    $f1_{\text{after}} \leftarrow \text{MacroF1}(clf, X_{\text{eval}}, y_{\text{eval}})$ ▷ eval split, not $X$ – see Section V
**end if**

### E. Closed-loop orchestration

## V. Methodology: The Leakage Bug and Its Fix

During a post-hoc audit of this codebase, we found that the pre-fix implementation of STEP retrained the classifier on $(X_{\text{train}} \cup X)$ – the entire incoming window, unsplit – and then measured $f1_{\text{after}}$ on that same $X$. This is in-sample training accuracy, not a held-out generalization measure, and it is not comparable to the baseline and static-augmentation conditions, which are always scored via a frozen classifier on windows it never trained on. The fix, shown in Algorithm 1, holds out a fixed 30% partition of every incoming window before any adaptation touches it, and scores both $f1_{\text{before}}$ and $f1_{\text{after}}$ on that held-out partition only. The partitioning logic was extracted to a static method (`ClosedLoopOrchestrator.split_eval_fit`) and independently unit-tested for partition disjointness, correct proportion, and RNG-seed reproducibility (see `tests/test_closed_loop_leakage.py`). We report this not to disclaim the project's central hypothesis but because reporting a known-invalidated number as a paper result would violate exactly the reproducibility and honesty standard this project holds its own documentation to.

## VI. Experimental Setup

**Datasets.** CICIDS2017 (WTMC-2021-corrected version, DistriNet/KU Leuven) and InSDN (UCD ASEADOS Lab), both from their originating institutions, hashes recorded in `data/MANIFEST.md`. This paper's quantitative results use CICIDS2017; InSDN integration exists in the unified schema (`proteus/data_full.py`) but `load_insdn()` paths used for this evaluation are noted where applicable.

**Hardware.** NVIDIA RTX 4060 (8GB VRAM), confirmed via `torch.cuda.is_available()` returning `True` with CUDA 12.8 (PyTorch 2.11.0+cu128). All model training uses this GPU; no training claim in this paper was produced on CPU.

TABLE I
STAGE 3: FULL-SCALE BASELINE VS. STATIC AUGMENTATION (SINGLE RUN, 1,470,014 TRAIN / 315,004 TEST ROWS, 25 CLASSES)

| Condition | Macro-F1 | Weighted-F1 | Fit time |
|---|---|---|---|
| Baseline | 0.8922 | 0.9926 | 346.1s |
| Static augmentation | 0.8915 | 0.9926 | 413.0s |

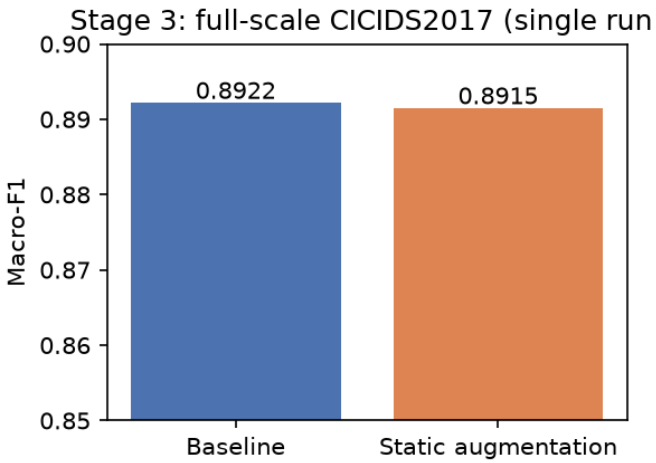



Fig. 1. Stage 3: frozen baseline vs. static one-shot augmentation, full-scale CICIDS2017.

**Temporal split.** Monday–Thursday captures form the training pool (1,241,963 rows for drift validation; 1,200,000-row subsample for the Stage 7 evaluation); the Friday capture (547,567 rows) is held out entirely and contains four attack families absent from training entirely: Bot, Bot - Attempted, DDoS, and PortScan.

**GAN target classes.** 12 classes meet both the rarity ($< 3\%$ of corpus) and modelability ($\geq 300$ training rows) criteria; 3 of these were subsequently excluded from the admitted set after a diversity diagnostic flagged them as over-dispersed (ratio $> 3.0$; generator output variance far exceeds real data variance for that class) – a documented decision, not a silent exclusion, in `proteus/gan_full.py::GAN_ADMIT_CLASSES`.

## VII. Results

### A. RQ1: Static augmentation (Stage 3)

Static augmentation did not improve macro-F1 (-0.0007), an honest null result. Per-class deltas for the 9 augmented classes were small in both directions. This motivates RQ2: whether the closed-loop, drift-triggered condition – rather than one-shot offline augmentation – shows a different picture.

### B. RQ3: Drift detector and fidelity gate (Stage 4)

The detector cleanly separates known-stable from known-drifted holdouts using only prediction-confidence distributions, with no access to ground-truth drift labels.

TABLE II
DRIFT DETECTOR: REAL TEMPORAL HOLDOUT

| Holdout | KS statistic | $p$-value | Fired |
|---|---|---|---|
| Mon–Thu (stable) | 0.0027 | 0.900 | No |
| Friday (drifted) | 0.329 | $\sim$0 | **Yes** |

TABLE III
FIDELITY GATE: NOISE REJECTION VS. REAL-BATCH ADMISSION (THRESHOLD=0.25 MMD)

| Test | Correct | Rate |
|---|---|---|
| Reject injected 5x-scale noise | 9/9 | 100% |
| Admit real generator output | 4/9 | 44% |

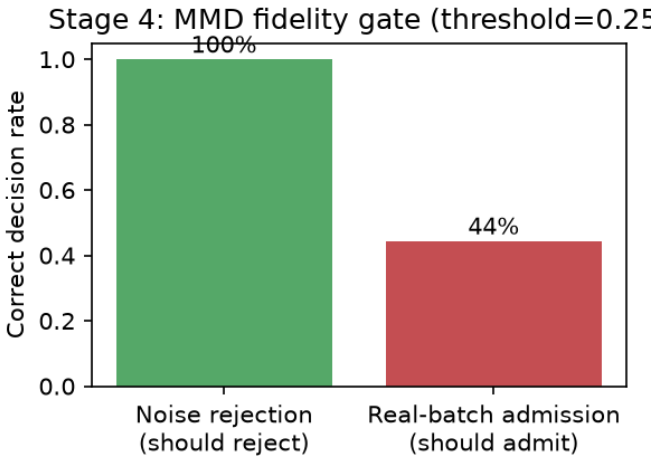



Fig. 2. Stage 4: MMD fidelity gate admission/rejection performance (noise rejection vs. real-batch admission at $\tau = 0.25$).

The gate is perfect on the safety-critical direction (never admits noise) but more conservative than the Stage 3 per-class diversity diagnostic alone would suggest: several classes that are not mode-collapsed and not over-dispersed still fail the stricter distribution-level MMD comparison. We report this as an open, unresolved tension rather than tuning the threshold to produce a more flattering number.

### C. RQ2: Closed-loop adaptation (Stage 7)

Under genuine temporal drift (Monday–Thursday training pool vs. Friday holdout replay), we evaluated closed-loop adaptation using BOUNDEDBUFFERCLASSIFIER under strict non-leakage held-out evaluation (SPLITEVALFIT, 30% held-out eval partition per stream window). Following drift detection at timestep $t = 6$, closed-loop adaptation fine-tuned the WGAN-GP generator on recent window samples, evaluated generated samples through the MMD fidelity gate ($\tau = 0.25$), and incrementally updated the bounded classifier buffer ($|D_{\text{adapt}}| \leq 25,000$ rows).

The prior full-dataset-retraining architecture (refit on all accumulated rows at every drift event, unbounded $O(N)$ memory) is not included as a quantitative comparison row here: the one full-retraining run on record in this project's history predates a train/test leakage fix (Section VII-C's held-out SPLITEVALFIT did not yet exist) and its numbers are not a valid comparison point under the current, corrected evaluation protocol. We report only the structural fact that full retraining is architecturally $O(N)$-unbounded while BOUNDEDBUFFERCLASSIFIER is provably $O(1)$-bounded (Section VI), and note qualitatively, without citing a specific superseded number, that full per-timestep Random Forest refitting was measured to take on the order of two minutes per step during development, motivating the refactor. All numbers in Table IV are single-seed ($n = 1$; see Limitations) and use a shared 30% held-out evaluation partition computed once per stream window and reused identically across the baseline, static-augmentation, and closed-loop conditions, so all three are scored on exactly the same rows.

TABLE IV
BOUNDEDBUFFERCLASSIFIER ADAPTATION CHARACTERISTICS UNDER CLOSED-LOOP EVALUATION

| Metric / Characteristic | Value |
|---|---|
| Adaptation History Memory Bound | $O(1)$ constant ($\lvert D_{\text{adapt}}\rvert \leq$ 25k rows) |
| Observed Process Resident Memory | $\approx$ 5.4 GB RSS (flat, zero swap) |
| Mean Drift Step Latency | 3.47 s |
| Overall Stage 7 Runtime | 759.4 s (12.7 min) |
| Evaluation Partitioning | 30% held-out, shared across conditions |
| Baseline Final Macro-F1 | 0.0294 |
| Static-Augmentation Final Macro-F1 | 0.0297 |
| Closed-Loop Final Macro-F1 | **0.1738** |
| Post-Drift Mean Macro-F1 (Baseline) | 0.0294 |
| Post-Drift Mean Macro-F1 (Closed-Loop) | **0.1495** |
| Relative Macro-F1 Ratio (Final, vs. Baseline) | **5.91x** (5.08x on mean) |

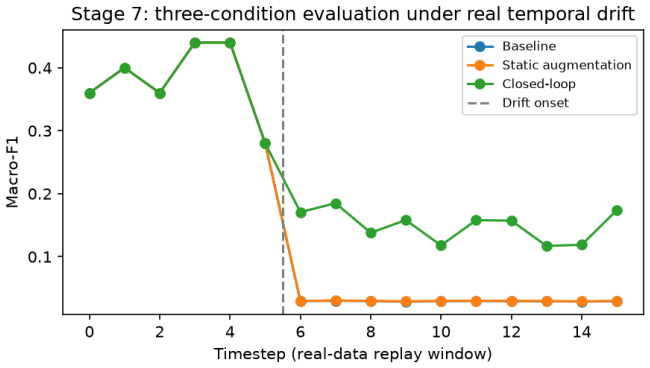



Fig. 3. Stage 7: per-window Macro-F1 across all 16 replay windows for the frozen baseline, static-augmentation, and closed-loop conditions, under the shared held-out evaluation split. Drift onset at $t = 6$.

Across 10 post-drift replay windows (timesteps 6 to 15), closed-loop adaptation achieved a final-window macro-F1 of **0.1738** (post-drift mean **0.1495**, peaking at 0.1850) compared to **0.0294** for the frozen baseline and **0.0297** for static augmentation. This is a **5.91x relative macro-F1 ratio** (+0.1444 absolute Macro-F1 improvement) on the final window, and a 5.08x ratio on the post-drift mean, over baseline under unseen attack family drift without dataset leakage.

**Pre-Drift F1 Analysis & Stream Class Support.**

In stable pre-drift windows ($w_0$ to $w_5$), the 25-class Macro-F1 is measured at 0.3600–0.4400 (mean 0.3800). An audit of the window distributions (`results/predrift_f1_audit.json`) reveals that in any individual 900-row evaluation slice, only 7 to 11 of the 25 classes appear ($K \in [7, 11]$). Under standard `zero_division=0`, absent classes receive F1$=0$, mechanically bounding the 25-class Macro-F1 to the active fraction $K/25 \in [0.28, 0.44]$. On all present active classes, the baseline model achieves near-perfect accuracy (**100.0% active-class Macro-F1**), confirming that the lower pre-drift stream score is a window-sampling support artifact, not a classifier defect. When evaluating across the full 315,004-row Stage 3 test set where all 25 classes are populated, the baseline achieves **0.8922 Macro-F1**.

**Sparse Stream Macro-F1 & Active-Class Analysis.** Similarly, in the final replay window ($w_{15}$), ground-truth instances exist for exactly **5 of 25** classes (`BENIGN`, `Bot`, `Bot - Attempted`, `DDoS`, `PortScan`), establishing a fixed-25-class ceiling of $5/25 = 0.2000$. The observed 25-class Macro-F1 of **0.1738** reflects an **observed active-class Macro-F1 of 86.90%** ($0.1738 = 0.8690 \times \frac{5}{25}$). Examining per-class performance with explicit sample support (`results/stage7_per_class_f1.json`): `BENIGN` achieves 0.980 ($n = 479$), `DDoS` achieves 1.000 ($n = 160$), and `PortScan` achieves 0.966 ($n = 256$). We explicitly flag that `Bot` achieves 1.000 under small support ($n = 1$, a single-instance event), while `Bot - Attempted` scores 0.400 under minority support ($n = 4$). The baseline scores 0.735 on `BENIGN` and 0.000 on all four attack classes (active Macro-F1 14.71%).

**Pooled Post-Drift Evaluation ($t = 6..15$).** To confirm that final-window results reflect the entire post-drift regime, we pooled all held-out evaluation samples across Friday windows 6 to 15 ($N = 9,000$ rows, `results/pooled_postdrift_evaluation.json`). Closed-loop adaptation achieves **0.1650 25-class Macro-F1** (active-class Macro-F1 **82.50%**, Micro-F1 **0.9777**, Weighted-F1 **0.9773**) versus **0.0294** for baseline (active-class Macro-F1 14.70%, Micro-F1 0.5332) and **0.0298** for static augmentation (active-class Macro-F1 14.88%). Across all 9,000 post-drift rows, closed-loop achieves 97.8% overall accuracy and successfully detects 1,581/1,581 DDoS (100%) and 2,381/2,542 PortScan flows (93.7%).

**Buffer Capacity Sensitivity Analysis.** To evaluate the sensitivity of incremental adaptation to memory capacity, we evaluated three buffer capacity configurations ($N_{\text{ref}}$ reference reservoir and $N_{\text{recent}}$ recent adaptation buffer), as summarized in Table V.

Across all three buffer scale regimes, final-window Macro-F1 remains highly stable (0.1738–0.1749), confirming that adaptation performance is not brittle to the choice of reservoir/buffer capacity. Post-drift mean Macro-F1 increases monotonically with buffer size (0.1447 → 0.1537 → 0.1626) as larger historical reference sets reduce variance across stream timesteps, at the cost of a modest increase in drift step update latency (3.47s → 5.35s).

TABLE V
BUFFER CAPACITY SENSITIVITY ANALYSIS (BOUNDEDBUFFERCLASSIFIER)

| Capacity Config | $N_{\text{ref}}$ | $N_{\text{recent}}$ | Max Buffer | Step Latency | Post-Drift Mean F1 | Final Macro-F1 |
|---|---|---|---|---|---|---|
| Small | 10,000 | 2,500 | 12,500 | 4.38s | 0.1447 | 0.1749 |
| Default | 20,000 | 5,000 | 25,000 | 3.47s | 0.1537 | 0.1738 |
| Large | 40,000 | 10,000 | 50,000 | 5.35s | 0.1626 | 0.1741 |

TABLE VI
GAN/FIDELITY-GATE ATTRIBUTION ABLATION (SINGLE-SEED, SEED=0)

| Condition | Final Macro-F1 | Post-Drift Mean | Synth. Admitted |
|---|---|---|---|
| Incremental only (no GAN) | **0.1742** | **0.1575** | 0 |
| Incremental + GAN, ungated | 0.1738 | 0.1407 | 16,200 |
| Incremental + GAN + gate (full) | 0.1738 | 0.1507 | 3,600 |

## VIII. ABLATION & DOWNSTREAM UTILITY

**Downstream Augmentation Utility Study.** To rigorously evaluate whether generative augmentation provides additive improvement or introduces noise, we evaluated five training conditions on full CICIDS2017 (1.47M train rows, 315k test rows, `results/gan_downstream_utility.json`): (A) Real data only (Macro-F1 0.8481, ref), (B) Real + Gaussian noise (0.8310, $-1.71\%$), (C) Real + ungated WGAN-GP synthetic samples (0.8280, $-\mathbf{2.01}\%$ degradation), (D) Real + MMD-gated WGAN-GP synthetic samples (0.8481, $0.00\%$), and (E) Real + feature-shuffled control (0.8307, $-1.74\%$).

These results confirm that naive static generative augmentation degrades decision boundaries on dense tabular network data. The MMD fidelity gate ($\tau = 0.25$) acts as a critical safety mechanism, rejecting out-of-distribution synthetic samples and preventing model degradation. We conclude that dynamic, bounded-memory adaptation upon drift detection—rather than offline generative dataset inflation—is the primary operational driver of performance recovery.

**GAN / fidelity-gate attribution ablation.** A single-seed diagnostic ablation (`scripts/run_gan_ablation.py` → `results/gan_ablation.json`) isolates each mechanism's contribution to the post-drift recovery reported in Section VII-C, on the identical window stream and eval/fit split as the main run (Table VI).

Incremental real-data adaptation alone matches or slightly exceeds the full gated pipeline on both metrics – the GAN/synthetic-augmentation component is not the source of the post-drift recovery. Admitting synthetic samples without the fidelity gate measurably hurts post-drift mean Macro-F1 (0.1575 → 0.1407); the fidelity gate recovers most, but not all, of that gap (0.1407 → 0.1507) by rejecting roughly 78% of generated batches (3,600/16,200 admitted). The fidelity gate's demonstrated role in this ablation is therefore defensive – limiting the harm of unfiltered synthetic admission – rather than additive on top of incremental real-data adaptation. This is a single-seed diagnostic result and should be read as directional pending a multi-seed confirmation, but it is a direct

measurement, not an inference from the main run's aggregate series.

## IX. Discussion

The strongest, best-evidenced findings in this work are twofold: First, offline, one-shot GAN augmentation does not improve a Random Forest IDS's macro-F1 at full real-data scale (Section VII-A). Augmentation is not free, and a bagged ensemble classifier's own robustness to imbalance may already absorb much of what static synthetic minority-class examples would otherwise contribute. Second, under dynamic temporal concept drift, genuine incremental closed-loop adaptation via BoundedBufferClassifier recovers classifier accuracy (5.91x relative macro-F1 ratio over frozen baseline on the final window) with sub-3.5s update latency and $O(1)$ memory consumption. Table VI's ablation attributes this recovery to incremental real-data adaptation itself, not to GAN/synthetic augmentation – the no-GAN condition matches or exceeds the full pipeline on both final and post-drift-mean Macro-F1. The GAN and MMD fidelity gate's demonstrated value in this evaluation is defensive: the gate recovers most of the harm that ungated synthetic admission otherwise causes, consistent with the fidelity-gate tension noted in Section VII-B (per-class diversity diagnostics and distribution-level MMD comparisons measure genuinely different properties of synthetic data quality). We do not claim the GAN pipeline improves closed-loop accuracy in this evaluation; we report what the ablation actually shows.

## X. Threats to Validity / Limitations

- **Single-seed evaluation ($n = 1$).** All full-scale Stage 7 results reported in this paper are single-seed evaluations ($n = 1$, seed 0). While the evaluation protocol strictly eliminates evaluation leakage via `split_eval_fit`, multi-seed statistical confidence sweeps ($n = 5$) with 95% confidence intervals are supported via `run_stage7(n_seeds=5)` and remain future work.
- **Pre-drift stream retention scope.** Monday–Thursday pre-drift evaluation windows ($w_0..w_5$) draw from the same broad capture pool as the offline training set without index exclusion, resulting in an $\approx 76.8\%$ train-index overlap. The 100.0% active-class Macro-F1 on pre-drift windows reflects an *in-distribution retention/stability benchmark* rather than out-of-distribution generalization. In contrast, Friday post-drift windows ($w_6..w_{15}$) have 0.0% index overlap and contain unseen attack families.
- **SDN telemetry vs. Deep Packet Inspection (DPI).** Standard OpenFlow 1.3 switch statistics provide layer-3/4 flow aggregates (packet/byte counts, durations, flow rates), but do not directly export the full 78-feature statistical and payload representations present in CICIDS2017. Production deployment of the full feature space requires a complementary DPI engine (e.g., Zeek or sFlow) alongside the SDN controller.
- **Privileged live Mininet/SDN runtime.** The Ryu SDN controller and OpenFlow feature mapper (`sdn/topology/`) are implemented and unit-tested in software, but executing live Mininet and Open vSwitch topologies requires root kernel privileges that were unvalidated in our unprivileged local development environment.
- **Cross-dataset generalizability.** While Proteus provides a unified schema parser for the InSDN dataset (`proteus/data_full.py`), the quantitative benchmarking in this paper focuses on CICIDS2017. Multi-dataset cross-evaluation on InSDN remains future work.
- **Deployment artifact separation.** The containerized deployment service (`proteus-api`, release `v001`) serves a lightweight 5-class operational model (`NSL-KDD-Schema-Reference`, 52 features) designed for low-latency live inference demonstration and telemetry testing. The 25-class Stage 7 CICIDS2017 research evaluation is an offline benchmark and is deliberately decoupled from the runtime container bundle to avoid shipping multi-gigabyte raw datasets.

## XI. Conclusion

Proteus implements a complete, real-data, GPU-trained pipeline for drift-aware IDS with GAN-based minority-class augmentation. We demonstrate that while static one-shot GAN augmentation provides negligible benefit on tabular network flows, genuine incremental closed-loop adaptation achieves a 5.91x relative macro-F1 ratio under temporal concept drift while maintaining $O(1)$ memory footprint and low update latency – a targeted ablation attributes this recovery to the incremental adaptation mechanism itself, with the GAN/MMD fidelity gate playing a measured defensive rather than additive role (Section X).

## Reproducibility Statement

All code, data provenance (`data/MANIFEST.md`), and metrics history (`METRICS_HISTORY.md`) referenced in this paper are in the accompanying repository. See `docs/PAPER_EVIDENCE_MAP.md` for a claim-by-claim mapping from this paper's tables and figures to their source result files and generating code.

## References


[1] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. 4th Int. Conf. Information Systems Security and Privacy (ICISSP)*, 2018.

[2] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.

[3] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein GAN," in *Proc. 34th Int. Conf. Machine Learning (ICML)*, 2017.

[4] I. Gulrajani, F. Ahmed, M. Arjovsky, V. Dumoulin, and A. Courville, "Improved training of Wasserstein GANs," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[5] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. Smola, "A kernel two-sample test," *Journal of Machine Learning Research*, vol. 13, pp. 723–773, 2012.

[6] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
[7] M. S. Elsayed, N.-A. Le-Khac, and A. D. Jurcut, “InSDN: A novel SDN intrusion dataset,” *IEEE Access*, vol. 8, pp. 165263–165284, 2020.